ECONOMIC INFORMATICS · BACHELOR · BUCHAREST UNIVERSITY OF ECONOMIC STUDIES

Multimedia

Lecture 1 — Concepts and Business Applications

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AGENDA

What we cover today

01	What multimedia is: definition, drivers, media types, interactivity
02	From data to media: how large images, sound and video are
03	Production chain, devices, authoring and the web platform
04	Multimedia applications in business and current trends
05	Course organisation, assessment and lab preparation

01 · DEFINITION

What multimedia is

Multi = many · **media** = means of communication

> The computer-controlled integration of text, graphics, images, sound, video and animation, where every medium is represented, stored, processed and transmitted digitally, and users interact with it through a digital device.

Digitised

Every medium is binary data that software can store, process and transmit.

Distributed

Content is stored remotely or produced live, and delivered over networks.

Interactive

Users influence what they receive, beyond play, pause and fast-forward.

Integrated

Media are orchestrated together, yet each can be handled on its own.

01 · DRIVERS

What drives multimedia

Compression

Codecs shrink media enough to store and stream them

Bandwidth

Broadband, fibre and 5G carry video to every user

Ubiquitous access

Wireless networks and the internet: anywhere, anytime

Processing and storage

Fast CPUs and GPUs, large SSDs, computer vision, speech recognition

Smart devices

Smartphones, tablets, smart TVs and wearables, all connected

Digitisation

Cameras, microphones and scanners; digitised text, audio and video archives

Static and time-based media

Static (discrete)

Independent of time: displayed once

Text — Unicode, HTML, PDF, hypertext

Still image — photos, scans, a grid of pixels

Graphics — computer-drawn shapes: logos, icons, charts

Time-based (continuous)

Depend on time: must be played back and synchronised

Audio — speech, music, effects, MIDI

Video — a sequence of frames, usually with sound

Animation — a sequence of generated images

Interactive 3D — scenes rendered in real time

Other criteria — **origin:** captured vs. synthesised (a photo vs. a MIDI track or a 3D render) · **storage:** standalone vs. stored in a network

Hypertext, hypermedia, hypervideo

Hypertext

Text read non-linearly, by following links to other parts or other documents.

Example: Wikipedia, online documentation

Hypermedia

Links between any media: images, graphics and continuous media such as sound and video.

Example: a product page with gallery, video and reviews

Hypervideo

The video stream itself holds interactive anchors that lead to other media.

Example: shoppable video, where clicking a jacket in the clip opens its product page

Linear or non-linear, passive or proactive

Linear: a fixed path from start to end (film, slideshow). **Non-linear:** the user chooses the path through the content (website, game, app) — hypermedia.

Level	What the user does	Business example
Passive	Watches; controls only playback	Video ad, product spot
Reactive	Chooses from predefined options	Catalogue with filters, satisfaction survey
Proactive	Creates or modifies the content	Product configurator, ad editor

02 · FROM DATA TO MEDIA

11.2 GB

is the size of a single minute of uncompressed Full HD video.

1920 × 1080 pixels × 3 bytes × 30 frames/s × 60 s

How large uncompressed media are

Media	Parameters	Calculation (bytes)	Result
Full HD image	1920 × 1080 px, 24 bits	1920 · 1080 · 3	6.2 MB
CD audio, 1 minute	44.1 kHz, 16 bits, stereo	44,100 · 2 · 2 · 60	10.6 MB
Full HD video, 1 s	30 frames/s	6,220,800 · 30	186.6 MB
Full HD video, 1 min	60 s	186,624,000 · 60	11.2 GB

Without compression, multimedia would be impractical both on disk and over networks. Lossless and lossy compression return in the lectures on images (3), sound (7) and video (9).

03 · PRODUCTION

The multimedia production chain

1

Concept

Business objective, target audience, script, storyboard

2

Capture and creation

Photography, recording, 3D modelling, AI generation

3

Processing

Retouching, editing, mixing, compression, format conversion

4

Integration

Assembly into an interactive application, on the web or in Unity

5

Delivery

Web publishing, streaming, app stores, content delivery networks (CDN)

6

Measurement

Views, watch time, conversions; adjusting the content

How devices digitise media

Analog signal
light, sound waves

→

Sensor
CCD or CMOS, microphone

→

ADC
analog-to-digital converter

→

DSP
adjusts quality, compresses

→

Storage
memory card, or RAM via USB

Scanner

Light reflected from the page reaches a CCD; OCR turns printed text into editable text.

Digital camera

The lens focuses light on the sensor; a DSP adjusts the image and saves it.

Sound card

An ADC digitises the microphone; a DAC converts back for the speakers.

Video

Video camera, webcam or capture card; the video card drives the display.

Playback reverses the chain: digital data → DAC → speakers or display.

The hardware and software we use

Hardware

Capture: scanner, camera, microphone, webcam

Processing: CPU, GPU, NPU for AI acceleration

Playback: sound and video cards, HDR display, VR/AR headsets

Storage and network: SSD, broadband, CDN

Software

Device drivers: let the operating system control capture and playback hardware

OS media utilities: player, image viewer, basic editor

Media-specific apps: GIMP, Audacity, DaVinci Resolve, Blender

Authoring and integration: Unity, web APIs in VS Code

Authoring and programming

Authoring: an integrated environment with visual interface design and high-level scripting. **Programming:** code and libraries give full control over every resource.

Approach	How content is organised	Best for	Classic tools	Today
Book metaphor	Media placed on pages, like a book	Kiosks, lectures, tutorials	ToolBook	PowerPoint, Genially
Timeline	Frames on a timeline, like a film	Precise sync, advanced animation	Director, Flash	Adobe Animate, Unity Timeline
Flow diagram	Icons on flow lines, for content and interactions	Tutorials, product demos, simulations	Authorware	Twine, visual scripting
Programming	Code with specialised libraries	Custom, high-performance applications	C, Java	JavaScript, C#

03 · TWO APPROACHES

Native web APIs and a production engine

Web: HTML, CSS, JavaScript

Fine control over every pixel and sample. Runs in any browser, with no installation.

Best for: websites, business applications, lightweight components.

Unity: editor and C#

Scenes, components, physics and animation ready to use. Exports to web, mobile, desktop and VR.

Best for: 3D configurators, simulations, games, immersive experiences.

During practical activities we build the same topic in both environments.

The browser as a complete multimedia platform

Media type	Element or API
Image	<img>, <picture>
Image processing	Canvas 2D
Vector graphics	SVG
Animation	CSS, Web Animations API
Sound	<audio>, Web Audio API
Video	<video>, WebCodecs
Capture and real time	getUserMedia, WebRTC
3D	WebGL, WebGPU

03 · THE WEB PLATFORM

One web stack, many targets

Browser

Client-side web apps, such as Gmail or online media players

Server

Node.js: media conversion, streaming servers, APIs

Desktop

Electron packages web apps as desktop apps, such as VS Code

Mobile

Apache Cordova or Capacitor wrap web apps; PWAs install directly

Beyond audio, video and graphics, web APIs add offline use, persistent storage, geolocation, device orientation and other sensors.

04

Multimedia applications in business

Where multimedia creates value for a company

04 · APPLICATION CLASSES

Six business domains

E-commerce

Product galleries, 360° views, product videos, 3D configurators

Digital marketing

Animated banners, short social media video, interactive ads

Corporate communication

Podcasts, webinars, interactive annual reports

E-learning and training

Video courses, simulations, employee training

Data visualisation

Animated dashboards, infographics, visual reports

Gamification

Loyalty programmes, branded games, augmented reality filters

From business scenario to media

Scenario	Media involved	Business value
Online furniture shop	3D, image, animation	Customers see the product in the chosen variant
Product launch	Video, audio, animation	A memorable message, shared on social media
A bank's podcast	Audio	Financial education, customer relationship
Interactive annual report	Vector graphics, animation	Complex data, easy to explore
Photo booth at an event	Real-time video, image	Participant-generated content, visibility

04 · TRENDS

What is changing in multimedia now

Generative AI	Images, voice and video generated or edited automatically; content provenance matters (C2PA)
GPU in the browser	WebGPU runs in Chrome, Edge, Firefox and Safari: 3D and AI with no installation
Short vertical video	A central format in social media marketing
Mandatory accessibility	The European Accessibility Act has applied since June 2025: captions, alternative text
Spatial media	WebXR, VR/AR headsets, virtual showrooms

05

Course organisation

Structure, labs, project and assessment

Objective and teaching staff

> Acquire the theoretical and practical knowledge needed to develop multimedia applications.

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COURSE PLATFORM

online.ase.ro

Slides, lab materials, assignments

05 · STRUCTURE

14 lectures, seven blocks

Weeks	Block	Main topics
1–2	Foundations	Concepts, business applications, Unity
3–6	Image and graphics	Raster, Canvas processing, SVG, animation
7–8	Sound	Digital audio, codecs, Web Audio API
9–10	Video and real time	Video codecs, streaming, WebCodecs, WebRTC
11	3D	Graphics pipeline, WebGPU, Unity
12–13	AI and delivery	In-browser AI, PWA, accessibility; project presentations in lab 13
14	Final assessment	Wrap-up; practical and theoretical final assessment

05 · ASSESSMENT

How the grade is formed

25%

Final: practical

On computer, week 14

30%

Final: theoretical

On paper, week 14

30%

During the semester

20% lecture + 10% labs: midterm, quizzes, tests, homework, participation

15%

Project

Presented in the lab, week 13

All components are mandatory. To pass, the final assessment, the project and the semester activity must each be graded at least 5 out of 10.

Cooperation, originality, creativity

Cooperation is welcome	Discussing the material with colleagues to understand it better is warmly encouraged
Your own work	Every assignment you hand in must be written by you
No plagiarism	Copying from others is condemned, and measures will be taken
Creativity	Original ideas and design choices are strongly encouraged
AI assistants	Allowed if declared in an AI.md file; you must be able to explain every line

SELF-ASSESSMENT

Review questions

1. Name the four key properties of a multimedia system.
2. Which media depend on time, and why do they raise synchronisation problems?
3. How does hypervideo differ from hypermedia?
4. How large are 30 seconds of uncompressed stereo audio at 44.1 kHz, 16 bits?
5. Describe the path from light to a stored file in a digital camera.
6. Compare the timeline and flow-diagram authoring metaphors.
7. Give a business example for each level of interactivity.

RESOURCES

Bibliography and resources

Books

Li Z.-N., Drew M. S., Liu J., *Fundamentals of Multimedia*, 3rd ed., Springer, 2021

Poynton C., *Digital Video and HD: Algorithms and Interfaces*, 2nd ed., Morgan Kaufmann, 2012

Shiffman D., *The Nature of Code*, No Starch Press, 2024

Marinacci J., *HTML Canvas Deep Dive*, joshondesign.com

Online

Course platform: online.ase.ro

References: MDN Web Docs, W3Schools, web.dev, Unity Learn

Browser support: caniuse.com

Code playgrounds: CodePen, JSFiddle, JS Bin

Media for projects: unsplash.com, pond5.com

Questions?